

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent

12388525

Kind Code

B1

Date of Patent

August 12, 2025

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Method and system for directly connecting a PON device to a mobile device to provide real-time information

Abstract

Techniques for repairing a network failure in a Passive Optical Network (PON) include establishing a communication session with an optical network terminal (ONT) in a PON via a short-range communication link. In response to establishing the communication session, the techniques include receiving diagnostic information related to a network failure corresponding to the ONT, and presenting the diagnostic information to a user for the user to repair the ONT based on the received diagnostic information.

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Appl. No.: 19/062499

Filed: February 25, 2025

Related U.S. Application Data

continuation parent-doc US 18889728 20240919 US 12267104 child-doc US 19062499

continuation parent-doc US 18636808 20240416 US 12136944 20241105 child-doc US 18889728

Publication Classification

Int. Cl.: H04B10/07 (20130101); H04B10/073 (20130101); H04B10/079 (20130101);
H04Q11/00 (20060101)

U.S. Cl.:

CPC **H04B10/073** (20130101); **H04B10/0793** (20130101); **H04Q11/0067** (20130101);
H04Q2011/0083 (20130101)

Field of Classification Search

CPC: H04B (10/07); H04B (10/079-0799)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) The present application is a continuation of U.S. application Ser. No. 18/889,728 filed Sep. 19, 2024, entitled “Method and System for Directly Connecting a PON Device to a Mobile Device to Provide Real-Time Information,” which is a continuation of U.S. application Ser. No. 18/636,808 filed Apr. 16, 2024, entitled “Method and System for Directly Connecting a PON Device to a Mobile Device to Provide Real-Time Information,” the disclosure of which is incorporated herein by reference in its entirety for all purposes.

FIELD OF THE DISCLOSURE

(1) This disclosure relates generally to passive optical networks (PONs), and, more particularly, to systems and methods for directly connecting to a device in the PON and presenting diagnostic information to assist in repairing the device.

BACKGROUND

(2) A conventional PON includes one or more optical line terminals (OLTs) at a central location connecting to one or more optical last mile termination units (LMTUs) disposed at respective customer premises (e.g., physical locations serviced by the PON) via one or more optical fibers. A PON is typically implemented using a point-to-multipoint topology in which a feeder optical fiber from an OLT serves multiple last mile termination units. An LMTU may be, for example, an optical network terminal (ONT) or an optical network unit (ONU) that is optically connected to the OLT via a respective distribution optical fiber received at the LMTU. Typically, the distribution optical fibers for respective ones of the LMTUs are optically coupled to the feeder optical fiber via a fiber distribution hub (FDH) using an optical splitter. A fiber distribution terminal (FDT) may be utilized to connect feeder optical fibers to distribution optical fibers, for example.

(3) When a PON experiences a network failure (e.g., equipment malfunctions, signal loss, network congestion, broken optical fibers, or other technical issues), the effects are typically observed by customers or end-users at locations at which last mile termination units are disposed. For example, a customer may observe a slowing down or lack of fidelity of PON services at his or her location, and may contact the PON service provider for assistance.

(4) However, a technician typically needs to have years of experience with troubleshooting to understand how to properly repair the network failure. Additionally, the technician may have difficulty locating the network equipment within a customer's residence. Moreover, it can be time consuming for the technician to correctly diagnose the problem before beginning the repair. Even if the technician is able to read information from the ONT, technicians may need years of training and experience before they can quickly identify the problem and resolve the issue.

SUMMARY

(5) To obtain diagnostic information for repairing an ONT, a user directly connects their client device to an ONT or another device in the PON (e.g., an FDT, an FDH, etc.) by establishing a communication session with the ONT via a short-range communication link. For example, the client device and the ONT may connect using Wi-Fi, Bluetooth, or near field communication (NFC). Then the ONT can provide real-time diagnostic information from the ONT to the client device to assist the user in diagnosing and repairing the failure. In some implementations, the ONT may run diagnostic tests and/or may receive requests from the client device to run diagnostic tests, such as a dial tone test to test whether the ONT can successfully complete a call with the client device or a speed test to test download or upload rates at the ONT. Additionally, the ONT may receive diagnostic information from other devices in the PON. For example, the ONT may receive diagnostic information from the FDT, FDH, and/or OLT optically connected to the ONT.

(6) In this manner, a user such as a technician may review the diagnostic information and identify the source of the network failure. Moreover, by directly connecting to the ONT, the ONT does not have to send information over the optical fibers to the OLT and to a server before it is sent to the client device. This can be particularly problematic when there is a network failure in the optical fibers or at another component of the PON. The server may not be able to receive the information needed to assess and diagnose the problem. By directly connecting the client device to the ONT via a short-range communication link, the client device can receive real-time diagnostic information even when data is not being received at the OLT from the ONT.

(7) Furthermore, the client device may store a geospatial tag indicating the location of the ONT. Then when the client device is within a threshold distance of the ONT, the client device may automatically connect to the ONT. Moreover, the client device may present a map display

indicating the location of the ONT relative to the user to direct the user to the ONT or to another component of the PON, such as the FDT or the FDH.

(8) In addition to diagnosing the problem, the client device may determine how to repair the problem and may provide step-by-step instructions to the user for repairing the problem. In this manner, the user does not necessarily have to be a technician to perform the repair. Instead, a customer can make their own repairs by following the step-by-step instructions. In some implementations, the client device includes a transparent display, such as a camera view depicting the area in front of the user in the camera view. For example, the client device may be a wearable device such as a virtual reality (VR) and/or augmented reality (AR) headset or smart glasses. When the user is facing the ONT, the transparent display may include a camera view of the ONT with the step-by-step instructions overlaid on the transparent display. The instructions may be presented in an area of the transparent display which does not obstruct the user's view of the ONT, so that the user can simultaneously view the instructions and the ONT to perform the repairs while reading the instructions.

(9) Additionally, the client device may overlay AR features on the transparent display to further guide the user when making the repair. For example, the client device may analyze images in the camera view to identify objects within the camera view, such as the ONT, optical fibers coupled to the ONT, the ports on the ONT, light emitting diodes (LEDs) on the ONT, etc. The client device may then overlay AR features to highlight certain components of the ONT corresponding to the instructions for repairing the ONT. For example, an instruction may be to check that a particular LED blinks three times. The client device may overlay an AR feature over the particular LED (e.g., a transparent yellow box) so that the user knows where to look to verify that the LED blinked three times. Another instruction may be to replace a particular optical fiber connected to the ONT. The client device may overlay an AR feature (e.g., a transparent green oval) over the particular fiber to highlight the particular fiber. In other implementations, the client device presents audio instructions via a speaker. In any event, the client device provides step-by-step instructions to the user based on diagnostics that the ONT obtained to assist the user in repairing the ONT. By overlaying AR features over portions of the camera view that correspond to the instructions, the client device makes it easier for the user to successfully perform the repair. This may increase the speed and the success rate of repairs for devices in a PON.

(10) If the ONT cannot be repaired and needs to be replaced, the client device may automatically order a new ONT to be delivered to the customer's residence. Then a user can install the new ONT using instructions provided by the client device and/or using the AR features overlaid on the transparent display to guide the user.

(11) In an embodiment, a method for repairing a network failure in a Passive Optical Network (PON) is provided. The method includes establishing a communication session, by a client device via a short-range communication link, with an optical network terminal (ONT) in a PON. In response to establishing the communication session, the method includes receiving, at the client device from the ONT via the short-range communication link, diagnostic information related to a network failure corresponding to the ONT. Additionally, the method includes presenting, by the client device, the diagnostic information to a user for the user to repair the ONT based on the received diagnostic information.

(12) In another embodiment, a client device for repairing a network failure in a Passive Optical Network (PON) includes one or more processors, and a non-transitory computer-readable memory storing instructions thereon. When executed by the one or more processors, the instructions cause the client device to establish, via a short-range communication link, a communication session with an optical network terminal (ONT) in a PON. In response to establishing the communication session, the instructions cause the client device to receive, from the ONT via the short-range communication link, diagnostic information related to a network failure corresponding to the ONT, and present the diagnostic information to a user for the user to repair the ONT based on the

received diagnostic information.

(13) In yet another embodiment, a non-transitory computer-readable memory stores instructions thereon. When executed by one or more processors, the instructions cause the one or more processors to establish, via a short-range communication link, a communication session with an optical network terminal (ONT) in a PON. In response to establishing the communication session, the instructions cause the one or more processors to receive, from the ONT via the short-range communication link, diagnostic information related to a network failure corresponding to the ONT, and present the diagnostic information to a user for the user to repair the ONT based on the received diagnostic information.

(14) In another embodiment, a method for augmenting repair of a network failure in a Passive Optical Network (PON) is included. The method includes receiving, via one or more sensors in a client device, sensor data indicating a current environment that includes an optical network terminal (ONT) in a PON. The PON includes an optical line terminal (OLT) optically connected to the ONT via one or more optical fibers. The method also includes analyzing, by the client device, the current environment to detect a cause of a network failure corresponding to the ONT, generating, by the client device, a set of instructions for repairing the ONT based on the detected cause of the network failure, and providing, by the client device, the set of instructions for a user to follow to repair the ONT. The set of instructions are provided as the user repairs the ONT.

(15) In yet another embodiment, a client device for augmenting repair of a network failure in a Passive Optical Network (PON) includes one or more sensors, one or more processors, and a non-transitory computer-readable memory storing instructions thereon. When executed by the one or more processors, the instructions cause the client device to receive, via the one or more sensors, sensor data indicating a current environment including an optical network terminal (ONT) in a PON. The PON includes an optical line terminal (OLT) optically connected to the ONT via one or more optical fibers. The instructions also cause the client device to analyze the current environment to generate a set of instructions for repairing the ONT, and provide the set of instructions for a user to follow to repair the ONT. The set of instructions are provided as the user repairs the ONT.

(16) In another embodiment, a non-transitory computer-readable memory stores instructions thereon. When executed by one or more processors, the instructions cause the one or more processors to receive, via one or more sensors, sensor data indicating a current environment including an optical network terminal (ONT) in a PON. The PON includes an optical line terminal (OLT) optically connected to the ONT via one or more optical fibers. The instructions also cause the one or more processors to analyze the current environment to generate a set of instructions for repairing the ONT, and provide the set of instructions for a user to follow to repair the ONT, wherein the set of instructions are provided as the user repairs the ONT.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying figures, where like reference numerals refer to identical or functionally similar elements throughout the separate views, together with the detailed description below, are incorporated in and form part of the specification, and serve to further illustrate examples of concepts that include the claimed invention, and explain various principles and advantages of those examples.

(2) FIG. 1 is a block diagram of an example passive optical network (PON) in which the systems, methods, and/or techniques of the present disclosure may be implemented.

(3) FIGS. 2A and 2B illustrate example displays of a PON repair application for connecting to an ONT using NFC, which may be presented on a client device.

(4) FIGS. 3A-3C illustrate example displays of a PON repair application for connecting to an ONT

by searching for the ONT and identifying a Wi-Fi signal broadcasted by the ONT, which may be presented on a client device.

(5) FIGS. 4A and 4B illustrate example displays of a PON repair application which include diagnostic information for the ONT and user controls for executing diagnostic tests on the ONT, which may be presented on a client device.

(6) FIG. 5 illustrates an example display of a PON repair application for executing a speed test for the ONT, which may be presented on a client device.

(7) FIGS. 6A and 6B illustrate example displays of a PON repair application for executing a dial tone test for the ONT, which may be presented on a client device.

(8) FIG. 7 illustrates an example transparent display of a client device with real-world imagery of the user's field of view which includes the ONT and step-by-step instructions for repairing the ONT overlaid on the transparent display.

(9) FIGS. 8 and 9 illustrate example transparent displays of the client device with real-world imagery of the user's field of view which includes the ONT and AR features overlaid on the transparent displays.

(10) FIG. 10 is a flowchart representative of an example method for repairing a network failure in a PON, which may be performed by a client device of FIG. 1.

(11) FIG. 11 is a flowchart representative of an example method for augmenting repair of a network failure in a PON, which may be performed by a client device of FIG. 1.

(12) Skilled artisans will appreciate that elements in the figures are illustrated for simplicity and clarity and have not necessarily been drawn to scale. For example, the dimensions of some of the elements in the figures may be exaggerated relative to other elements to help to improve understanding of the present disclosure.

(13) The apparatus and method components have been represented where appropriate by conventional symbols in the drawings, showing only those specific details that are pertinent to understanding examples of the present disclosure so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein.

(14) Although the figures show parts with clean lines and boundaries, some or all of these lines and/or boundaries may be idealized. In reality, the boundaries and/or lines may be unobservable, blended, and/or irregular. Use of terms such as up, down, top, bottom, side, end, front, back, etc. herein are used with reference to a currently considered or illustrated orientation. If they are considered with respect to another orientation, it should be understood that such terms must be correspondingly modified.

DETAILED DESCRIPTION

(15) Disclosed examples of the disclosure provide a number of advantages over existing techniques for repairing a PON. Reference will now be made in detail to non-limiting examples, some of which are illustrated in the accompanying drawings.

(16) Example Passive Optical Network (PON)

(17) FIG. 1 is a block diagram of an example PON **100** in which the systems, methods, and techniques of the present disclosure may be implemented. The example PON **100** includes one or more optical line terminals (OLTs) (an example one of which is designated by reference numeral **102**) at a central location (e.g., at a central office **104**) optically connecting to one or more last mile termination units **106a**, . . . , **106n** at respective customer premises **108a**, . . . , **108n**. The last mile termination units **106a**, . . . , **106n** may be located outside and/or inside the customer premises or locations **108a**, . . . , **108n**. Each last mile termination unit **106a**, . . . , **106n** may be, for example, an optical network unit (ONU) or an optical network terminal (ONT). In some examples herein, the term “optical terminal” generally refers to a last mile termination unit (e.g., an ONU or ONT) or an OLT.

(18) The example PON **100** is implemented using instances of point-to-multipoint topology. For

example, in the example PON **100**, a first feeder optical fiber **110a** from the OLT **102** (which is interchangeably referred to herein as an “F1 optical fiber **110a**” or a “primary optical fiber **110a**”) serves the one or more last mile termination units **106a**, . . . , **106n** via respective distribution optical fibers **112a**, . . . , **112n** (which are interchangeably referred to herein as “F2 optical fibers **112a**, . . . , **112n**” or “secondary optical fibers **112a**, . . . , **112n**”). In the illustrated example, the first feeder optical fiber **110a** is optically coupled to the plurality of last mile termination units **106a**, . . . , **106n** via an example one-to-many optical splitter **114a** which is disposed, located, implemented, etc. in an example fiber distribution hub (FDH) **116a**. In some arrangements, the FDH **116a** is located within a geographic area (e.g., a neighborhood) such that the customer premises **108a**, . . . , **108n** are proximally close to the FDH **116a**, and typically each of the customer premises **108a**, . . . , **108n** and respective last mile termination units **106a**, . . . , **106n** is disposed at a different optical distance from the FDH **116a**. An “optical distance,” as generally utilized herein, refers to a distance over which an optical signal travels.

(19) In embodiments, the PON **100** may or may not include additional feeder optical fibers and optical splitters for a plurality of additional customer premises. Moreover, a PON may or may not include a plurality of FDHs. For example, as shown in FIG. **1**, the example PON **100** includes a second feeder or primary optical fiber **110b** from the OLT **102** that is optically coupled to another plurality of last mile termination units **107a-107m** at respective customer premises **109a-109m** via another many-to-one optical splitter **114b** included in another fiber distribution hub **116b** and via respective secondary optical fibers **113a-113m**.

(20) As utilized herein, the “components” of the PON **100** generally refer to the devices, nodes, and optical fibers of the PON **100**. For example, the components of the PON **100** shown in FIG. **1** may include the OLT **102**, the FDHs **116a**, **116b**, the splitters **114a**, **114b**, the LMTUs **106a-106n** and **107a-107m**, and the optical fibers interconnecting the devices or nodes, e.g., the optical fibers **110a-110b**, **112a-112n**, and **113a-113m**.

(21) In some scenarios, an optical terminal (e.g., the OLT **102** and/or one or more the last mile termination units **106a-106n**, **107a-107m**) communicates with a computing device. For example, the OLT **102** and/or the one or more LMTUs **106a-106n**, **107a-107m** may transmit or receive data from a computing device **125** (e.g., a laptop, a computer, a tablet, a mobile phone, etc.) associated with a service technician or other agent of the PON **100** or a computing device associated with a customer.

(22) In some examples, the computing device **125** communicates with an LMTU **106a-106n**, **107a-107m** via one or more networks **128** (which may include one or more wired and/or wireless private networks and/or public networks, such as the Internet). In other examples, a client computing device **150** (also referred to herein as a “client device”) communicates directly with the LMTU **106a-106n**, **107a-107m** by establishing a communication session with the LMTU **106a-106n**, **107a-107m** over a short-range communication link, such as Bluetooth, Wi-Fi, or near field communication (NFC). The client device **150** may also be associated with a service technician, other agent of the PON **100**, or a customer. Then during the communication session, the client device **150** and the LMTU **106a-106n**, **107a-107m** may directly transmit information back and forth to each other. For example, the LMTU **106a-106n**, **107a-107m** may transmit diagnostic information to the client device **150** which may be presented on a user interface of the client device **150** via a PON repair application. The client device **150** may transmit control signals to the LMTU **106a-106n**, **107a-107m** for the LMTU **106a-106n**, **107a-107m** to perform diagnostic tests, such as a dial tone test or a speed test.

(23) The client device **150** may be a portable device such as a smart phone or a tablet computer, for example. The client device **150** may also be a laptop computer, a desktop computer, a personal digital assistant (PDA), a wearable device such as a smart watch or smart glasses, a virtual reality headset, etc.

(24) The client device **150** may include one or more processor(s) and a memory storing machine-

readable instructions executable on the processor(s). The processor(s) may include one or more general-purpose processors (e.g., CPUs), and/or special-purpose processing units (e.g., graphical processing units (GPUs)). The memory may be a non-transitory memory and can include one or several suitable memory modules, such as random access memory (RAM), read-only memory (ROM), flash memory, other types of persistent memory, etc. The memory may store instructions for implementing a PON repair application for directly connecting to an LMTU **106a-106n**, **107a-107m** and establishing a communication session with the LMTU **106a-106n**, **107a-107m** to receive diagnostic information for identifying the cause of a network failure. The PON repair application may also provide instructions for repairing the LMTU **106a-106n**, **107a-107m** and/or may present other graphical indications to guide a user through a repair.

(25) Additionally, the client device **150** may include sensors, such as a positioning sensor (e.g., a Global Positioning System (GPS)), an accelerometer, an infrared sensor, a camera, a depth sensor such as a light detection and ranging (lidar) sensor, an electromagnetic sensor, etc. The client device **150** may also include a network interface for communicating with devices over short-range communication links, such as Bluetooth, Wi-Fi, or NFC. The network interface may enable communication with other devices (e.g., the OLT **102**, the last mile termination units **106a**, **106n**, the one or more servers **130**, the computing device **125**, etc.) via any suitable networks such as the PON **100** and/or the network(s) **128**. The example network interfaces include any suitable type of communication interface(s) (e.g., wired and/or wireless interfaces) configured to operate in accordance with any suitable communication protocol(s). Example network interfaces include a TCP/IP interface, a WiFi™ transceiver (e.g., according to the IEEE 802.11x family of standards), an Ethernet transceiver, a cellular transceiver, a satellite transceiver, an asynchronous transfer mode (ATM) transceiver, a digital subscriber line (DSL) modem, a coaxial cable modem, a dialup modem, or any other suitable interface based on any other suitable communication protocols or standards.

(26) Furthermore, the client device **150** may include a display. The display may be a transparent display with camera views of real-world imagery. For example, when the client device **150** is smart glasses or a virtual reality headset, the transparent display may be presented in front of the user's eyes. Then the client device **150** may present augmented reality features on the transparent display overlaying the real-world imagery to help guide the user.

(27) Additionally and/or alternatively, the client device **150** or an optical terminal in the PON **100** may communicate with one or more servers **130** of the PON **100** that are used to manage the PON **100**, the network(s) **128**, etc. For example, the one or more servers **130** may schedule and execute diagnostics of various components of the PON **100** and/or of the PON **100** as a whole, generate alerts and alarms, initiate various actions, provide user interfaces, which may include graphical user interfaces (e.g., at the computing device **125**), log, historize, and/or otherwise store data generated by and associated with the PON **100** (e.g., in one or more data stores **132**), and the like. For example, one or more applications may execute at the server(s) **130** and/or the server(s) may host one or more services to provide management, administrative, and/or test functionalities of the PON **100**.

(28) In another example, the PON repair application may communicate with the server(s) **130** to transmit diagnostic information to the server(s), receive repair instructions from the server(s) **130**, receive a description of an identified root cause of the network failure from the server(s), etc. In some implementations, the entire functionality of the PON repair application may be executed on the server(s) **130**. In other implementations, the PON repair application does not communicate with a server **130** and performs its functionality locally on the client device **150**. In yet other implementations, the client device **150** and the server(s) **130** each perform a portion of functionality of the PON repair application.

(29) Various information and data associated with, utilized by, and/or generated by the PON **100** may be stored in the data stores **132** of the PON **100**. For example, the data store(s) **132** may store

sets of instructions for repairing LMTUs **106a-106n**, **107a-107m**, where each set of instructions corresponds to a different root cause of the network failure. For example, if the root cause is a broken optical fiber **112a**, the corresponding instructions may be for replacing the optical fiber **112a**. If the root cause is that the optical fiber **112a** is dirty, the corresponding instructions may be for cleaning the optical fiber **112a**. The data store(s) **132** may also store records of customer contact events with a technical support organization supporting the PON **100**, service call records, records of operating conditions and events which occurred, logbooks, and the like.

(30) Additionally, the data store(s) **132** may store applications which may execute at the one or more servers **130**, and/or which may be downloaded or otherwise provided to the technician computing device **125** for installation and execution thereon. Further, the data store(s) **132** may store data indicative of performance, faults, diagnostics, statuses, states, and/or other data corresponding to the components of the system **100**. Still further, the data store(s) **132** may store data indicative of the architecture, infrastructure, and component connectivity of the PON **100**, including identifications of various PON components and indications of which PON components connect to which other PON components. Of course, the data store(s) **132** may store any updates to any and all of the information and data stored therein.

(31) The example servers **130**, optical terminals (e.g., any of the OLT **102**, the last mile termination units **106a**, **106n**), and/or the example computing devices **125**, **150** may include a processing platform capable of executing instructions to, for example, implement operations of the example methods described herein, as may be represented by the flowcharts of the drawings that accompany this description. Other example logic circuits capable of, for example, implementing operations of the example methods described herein include a field programmable gate array (FPGA), an application specific integrated circuit (ASIC), a programmable logic device (PLD), and a field programmable logic device (FPLD). The processing platform may be, for example, one or more servers, a cloud computing system, a computer, a workstation, a laptop, a mobile device (e.g., a cell phone, a smart phone, a tablet such as an IPAD™), or any other type of computing device or system.

(32) The example processing platform includes one or more processors, one or more memories, one or more network interfaces, one or more input/output (I/O) interfaces, and/or a set of data stores, all of which are interconnected via one or more address/data bus or communication links.

(33) The processors may be implemented using hardware, and may include a semiconductor based (e.g., silicon-based) device. The processors may be, for example, one or more programmable microprocessors, controllers, digital signal processors (DSP), graphics processing units (GPU) and/or any suitable type of programmable processor capable of executing instructions to, for example, implement operations of the example methods described herein. Additionally and/or alternatively, the processors may be a field programmable gate array (FPGA), an application specific integrated circuit (ASIC), etc. that implements operations of the example methods described herein without executing instructions.

(34) The memories are accessible by the processors (e.g., via a memory controller). The example processors interact with the memories to obtain, for example, machine-readable instructions stored in the memories corresponding to, for example, the operations represented by the flowcharts of this disclosure. The example processors may also interact with the memories to store data, such as data formed or used during execution of machine-readable instructions. Example memories include any number and/or type(s) of volatile or non-volatile, non-transitory, machine-readable storage medium, devices or disks, such as a semiconductor memory, magnetically readable memory, optically readable memory, biologically readable memory, a hard disk drive (HDD), a solid-state drive (SSD), a read-only memory (ROM), a random-access memory (RAM), a redundant array of independent disks (RAID) system, a cache, flash memory, or any other storage medium, device or disk in which information may be stored for any duration (e.g., permanently, for an extended time period, for a brief instance, for temporarily buffering, for caching of the information, etc.).

Additionally and/or alternatively, machine-readable instructions corresponding to the example operations described herein may be stored on one or more volatile or non-volatile, non-transitory, machine-readable removable storage media (e.g., a compact disc (CD), digital versatile disk (DVD), Blu-ray disk, removable flash memory, etc.) that may be coupled to the processing platform to provide access to the machine-readable instructions stored thereon.

(35) The example processing platform includes one or more communication interfaces such as, for example, the one or more network interfaces, and/or the one or more input/output (I/O) interfaces. The communication interface(s) enable the processing platform to communicate with, for example, another device, system, etc. (e.g., the OLT **102**, the last mile termination units **106a**, **106n**, the one or more servers **130**, the computing device **125**, the one or more data stores **132**, etc., any other database, and/or any other machine).

(36) The example processing platform includes the network interface(s) to enable communication with other machines (e.g., the OLT **102**, the last mile termination units **106a**, **106n**, the one or more servers **130**, the computing device **125**, etc.) via, for example, one or more networks such as the PON **100** and/or the network(s) **128**. The example network interfaces include any suitable type of communication interface(s) (e.g., wired and/or wireless interfaces) configured to operate in accordance with any suitable communication protocol(s). Example network interfaces include a TCP/IP interface, a WiFi™ transceiver (e.g., according to the IEEE 802.11x family of standards), an Ethernet transceiver, a cellular transceiver, a satellite transceiver, an asynchronous transfer mode (ATM) transceiver, a digital subscriber line (DSL) modem, a coaxial cable modem, a dialup modem, or any other suitable interface based on any other suitable communication protocols or standards.

(37) The example, processing platform includes the input/output (I/O) interface(s) (e.g., a Bluetooth® interface, a near-field communication (NFC) interface, a universal serial bus (USB) interface, a serial interface, an infrared interface, a PCI express interface, etc.) to enable the processors to communicate with peripheral I/O devices and/or other communication systems. For example, the I/O interface(s) may be used to control a light source, enable receipt of user input (e.g., from a touch screen, a keyboard, a navigation device such as mouse, touch pad, joystick or trackball, a microphone, a button, etc.) and communicate output data (e.g., visual indicators, instructions, data, images, etc.) to the user (e.g., via a display, a speaker, a printer, a communication interface, an antenna, etc.). The I/O interface(s) typically include a graphics driver card, graphics driver chip and/or graphics driver processor to drive a display when a display is present.

(38) In some examples, the processing platform also includes, or is otherwise communicatively coupled to, a set of data stores or other data storage mechanisms (one or more of a HDD, optical storage drive, solid state storage device, CD, CD-ROM, DVD, Blu-ray disk, RAID, data storage bank, etc.). The set of data stores may include the example data stores **132**.

(39) To connect to an LMTU **106a** to establish a communication session between the client device **150** and the LMTU **106a**, a user may tap the client device **150** to the LMTU **106a** to cause the PON repair application to transmit an NFC signal to the LMTU **106a** requesting to pair the devices. The LMTU **106a** may receive the NFC signal and establish a communication session between the two devices **150**, **106a** in response to the request. Then the client device **150** and the LMTU **106a** may transmit NFC communications back and forth to each other with real-time information, such as diagnostic information from the LMTU **106a**, and control signals from the client device **150** for the LMTU **106a** to execute diagnostic tests.

(40) In another example, the client device **150** and the LMTU **106a** may connect via Wi-Fi. For example, the LMTU **106a** may broadcast a Wi-Fi signal. When the client device **150** is within communication range of the LMTU **106a**, the PON repair application may receive the Wi-Fi signal and may request to connect to the LMTU **106a** via the broadcasted Wi-Fi network. In some implementations, the PON repair application may provide a password for connecting to the LMTU **106a**. The LMTU **106a** may receive the request and establish a communication session between the

two devices **150**, **106a**. In some implementations, the LMTU **106a** may authenticate the client device **150** using the received password before establishing the communication session. Then the client device **150** and the LMTU **106a** may transmit Wi-Fi communications back and forth to each other with real-time information, such as diagnostic information from the LMTU **106a**, and control signals from the client device **150** for the LMTU **106a** to execute diagnostic tests.

(41) In some implementations, the PON repair application may store a geospatial tag indicating the location of the LMTU **106a**. For example, the PON repair application may determine the location of the LMTU **106a** based on the GPS location of the client device **150** while the client device **150** is connected to the LMTU **106a**. Then when the user is at the same location at a later time, the PON repair application may automatically connect to the Wi-Fi for the LMTU **106a**. In another example, the PON repair application may determine the location of the LMTU **106a** based on electromagnetic (EM) data detected from the client device **150** while the client device **150** is connected to the LMTU **106a**. The PON repair application may compare the EM data to known EM data at various locations to determine the location of the LMTU **106a**. Still further, if the user has trouble finding the LMTU **106a**, the PON repair application may use the stored geospatial tag to direct the user to the LMTU **106a**.

(42) If the user is within communication range of the LMTU **106a**, the PON repair application may precisely locate the LMTU **106a** relative to the user's location based on properties of the Wi-Fi signals communicated between the devices. For example, the PON repair application may determine the round trip time (RTT) it takes to transmit a message to the LMTU **106a** and receive a response message back from the LMTU **106a**. The PON repair application can determine the distance from the devices **150**, **106a** based on the round trip time for the communications back and forth. For example, the PON repair application can calculate the distance, D , as $\frac{1}{2} \times c \times \text{RTT}$, where c is the speed of light. Then the PON repair application may present an indication of the distance to the LMTU **106a** to the user to direct the user to the LMTU **106a**.

(43) The PON repair application may also determine the direction in which the user needs to travel to locate the LMTU **106a** based on the angle of arrival of communications between the client device **150** and the LMTU **106a**. For example, the client device **150** may have multiple antennas for receiving short-range communications. The client device **150** may receive a communication from the LMTU **106a** at each of its antennas, where the antennas are located at different positions within the client device **150**. The PON repair application may then determine the angle of arrival of the communication based on a time difference at which each of the antennas received the communication. In other implementations, the LMTU **106a** may have multiple antennas for receiving short-range communications and may determine the angle of arrival of a communication based on a time difference at which each of the antennas received the communication. In any event, the PON repair application may present an indication of the direction in which the user needs to travel to locate the LMTU **106a** to direct the user to the LMTU **106a**. For example, the PON repair application may present a map display with an indication of the location of the LMTU **106a** and an indication of the location of the user relative to the LMTU **106a**. The PON repair application may also provide navigation directions for traveling to the LMTU **106a**, such as "Go downstairs into the basement and the LMTU **106a** is to your left."

(44) In other implementations, the PON repair application may determine the direction in which the user needs to travel to locate the LMTU **106a** based on the EM data and/or accelerometer data. For example, the PON repair application may determine a change in position of the user based on the accelerometer data and/or a change in the EM data. Then the PON repair application may determine the direction in which the user is currently traveling based on the change in position and can provide directions to the LMTU **106a** using the user's current location and their direction of travel.

(45) In some implementations, the PON repair application may also connect to and/or receive navigation information to other optical terminals in the PON, such as an FDH **116a**, FDT, or OLT

102.

(46) FIGS. 2A-3C illustrate example displays **200-360** for connecting to an ONT or other optical terminal which may be presented by the PON repair application on the client device **150**. FIGS. 4A-6B illustrate example displays **400-650** for presenting diagnostic information and executing diagnostic tests which may be presented by the PON repair application on the client device **150**. FIGS. 7-9 illustrate example displays **700-900** for providing instructions for repairing an ONT or other optical terminal which may be presented by the PON repair application on the client device **150**. The screenshots shown in FIGS. 2A-9 are examples of what may be displayed to a user. As will be appreciated by those of ordinary skill in the relevant art(s), the exemplary screenshots shown in FIGS. 2A-9 are for illustrative purposes, and their associated functionality may be implemented using any suitable format and/or design for facilitating their corresponding described functionalities without departing from the spirit and scope of the present disclosure.

(47) As shown in FIG. 2A, the PON repair application may present a display **200** with instructions for connecting to an ONT **106a** using NFC. The display **200** may include instructions for connecting to the ONT **106a** by tapping the client device **150** to an NFC chip on the ONT **106a**. Then once the user taps the client device **150** to the ONT **106a**, the PON repair application may present a display **250** indicating the client device **150** is connecting to the ONT **106a**. The client device **150** may then pair with the ONT **106a** and establish a communication session between the devices **150**, **106a**.

(48) Additionally or alternatively, the client device **150** may connect to the ONT **106a** using a Wi-Fi communication link. As shown in FIG. 3A, the PON repair application may present a search display **300** for searching for a Wi-Fi network for connecting to the ONT **106a** by entering a customer identifier, an ONT serial number, or any other suitable identification information for the ONT **106a**. A user such as a customer or a technician may enter the customer identifier or the ONT serial number in a search bar and select a user control to submit the search query. The PON repair application may then identify the ONT **106a** based on the search query, and identify a Wi-Fi network for connecting to the ONT **106a**. FIG. 3B illustrates an example search result display **330** indicating the Wi-Fi network **332** for connecting to the ONT **106a** related to the search query and including a user control **334** for connecting to the Wi-Fi network. In response to receiving a selection of the user control **334** to connect to the Wi-Fi network **332**, the PON repair application may automatically connect to the Wi-Fi network **332** when the client device **150** is within communication range of the Wi-Fi network. For example, once the client device **150** detects a Wi-Fi signal from the Wi-Fi network **332**, the client device **150** may automatically connect to the Wi-Fi network **332**.

(49) In addition to entering identification information for the ONT **106a** in a search bar to identify the short-range communication network for connecting to the ONT **106a**, the ONT **106a** may include a barcode, such as a quick response (QR) code on the surface of the ONT **106a** with information for connecting to the short-range communication network. The client device **150** may capture an image of the barcode and analyze the image to obtain the information included in the barcode which may include the name of the Wi-Fi network, a password for connecting to the Wi-Fi network, etc.

(50) Then once the client device **150** connects to the ONT **106a** via a short-range communication link, the PON repair application may receive and present diagnostic information from the ONT **106a** as shown in the diagnostic display **360** of FIG. 3C.

(51) FIGS. 4A and 4B illustrate example diagnostic displays **400**, **450** which may be presented by the PON repair application on the client device **150**. For example, once the client device **150** connects to the ONT **106a** via a short-range communication link, the PON repair application may receive real-time diagnostic information from the ONT **106a**. If the client device **150** is not directly connected to the ONT **106a**, the client device **150** may receive stored diagnostic information from the server **130** which may not be up to date. In any event, the diagnostic display **400** includes a

graphic representation **402** of the ONT **106a** and an indication of the operational status **404** of the ONT **106a**. For example, the operational status **404** may indicate the ONT **106a** is operational or is experiencing a failure and is unable to communicate with the OLT **102**.

(52) The diagnostic display **400** also includes an indication of statuses **406** of neighbors' ONTs. In this manner, the user may be able to determine that the ONT **106a** is not the cause of the network failure and instead a neighbor's ONT is the cause of the network failure. Then the user may contact the neighbor to repair the neighbor's ONT.

(53) Additionally, the diagnostic display **400** includes a user control **408** for selecting an optical terminal in which to present diagnostic information. For example, the user can select to view diagnostic information for the ONT **106a**, the OLT **102**, the FDH **116a**, the FDT, etc. The ONT **106a** may receive diagnostic information from the OLT **102**, the FDH **116a**, the FDT, and/or any other optical terminal and transmit the diagnostic information to the client device **150** via the short-range communication link. When the user control **408** is selected for viewing diagnostic information for the ONT **106a**, the diagnostic display **400** presents a graphic representation **410** indicating the light level at the ONT **106a**. The graphic representation **410** may include an indication of the ONT light level (-14.0 dBm), an indication of a minimum and maximum light level (-27.696 dBm, -12.558 dBm) for all of the ONTs on the PON, and an indication of an average light level (-14.56 dBm) for all of the ONTs on the PON. Additionally, the graphic representation **410** may include a light level meter gauge having green, yellow, and red color indicators to indicate good, fair, and poor light level ranges, respectively. Still further, the diagnostic display may include a graphic indicator **412** below the light level meter gauge **410** indicating the light level range of all ONTs within a threshold radius (e.g., 1 km) of the ONT **106a**.

(54) Moreover, the diagnostic display **400** may include an indication of the distance **414** from the ONT **106a** to the OLT **102** (3.4 km) and an indication of the optical temperature **416** of the ONT **106a** (Normal).

(55) FIG. 4B illustrates another example diagnostic display **450** which may be presented by the PON repair application on the client device **150**. For example, the diagnostic display **450** may be presented when the user scrolls down on the diagnostic display **400** of FIG. 4A. In other implementations, the diagnostic display **450** may be presented in response to receiving a selection of a user control on the diagnostic display **400** to view another page.

(56) In any event, the diagnostic display **450** includes additional information about the optical temperature of the ONT **106a**. More specifically, the additional information includes a temperature metric **452** (32.0°) and a graphic representation of a temperature gauge **454** having green, yellow, and red color indicators to indicate normal, hot, and very hot temperature ranges, respectively. Furthermore, the diagnostic display **450** includes utilization metrics for the ONT **106a**. The utilization metrics include a PON utilization metric **456** (1%) and a network terminal (NT) uplink utilization metric **458** (10%).

(57) Additionally, the diagnostic display **450** includes user controls **460**, **462** for transmitting control signals to the ONT **106a** to cause the ONT **106a** to initiate diagnostic tests including a dial tone test **460** and a speed test **462**. The client device **150** may then receive diagnostic information from the ONT **106a** via the short-range communication link in response to the ONT **106a** performing the diagnostic test. In response to selecting the user control **462** to initiate a speed test, the ONT **106a** may receive the control signal from the client device **150** and begin executing a speed test to test the upload and download rates of the ONT **106a**. Then the ONT **106a** may transmit the results of the speed test to the client device **150** via the short-range communication link. When the user selects the user control **462** to initiate the speed test, the PON repair application may present a speed test display **500**, as shown in FIG. 5.

(58) The speed test display **500** may include a user control **502** for selecting a server for performing the speed test. Then the ONT **106a** may perform the speed test by transmitting and receiving data to and from the selected server over the fiber optic network and determining upload and download

rates of the communications with the selected server. The ONT **106a** may provide the upload and download rates to the client device **150** via the short-range communication link. Then the speed test display **500** may present indications **504**, **506** of the download and upload rates, respectively.

(59) In response to selecting the user control **460** to initiate a dial tone test, the ONT **106a** may receive a control signal from the client device **150** via the short-range communication link to call the client device **150**. The client device **150** may also transmit the client device's phone number to the ONT **106a**. Then the ONT **106a** may begin executing the dial tone test to test whether the ONT **106a** can successfully complete the call with the client device **150**. Then the ONT **106a** may call the client device **150** via the fiber optic network. When the user selects the user control **460** to initiate the dial tone test, the PON repair application may present a dial tone test display **600**, as shown in FIG. **6A**.

(60) The dial tone test display **600** may include an indication that the user should expect a call back shortly. Then as shown in FIG. **6B**, the client device **150** presents an indication of an incoming call for the user to accept with user controls for accepting or denying the call. The user can then determine that the ONT **106a** passed the dial tone test by being able to initiate a call with the client device **150**.

(61) The user may use the diagnostic information from the diagnostic tests and/or from the displays **400-650** to determine the root cause of the network failure and repair or replace the ONT **106a** or another device in the PON **100**. For example, in some scenarios, the user may determine that the ONT **106a** needs to be replaced and may swap the ONT **106a** out with a replacement ONT. The user may also order a new ONT to be shipped to the user's residence, for example if the user is a customer and not a technician and does not have a replacement ONT.

(62) In any event, the ONT **106a** may transmit any suitable diagnostic information to the client device **150** via the short-range communication link which may assist the user in determining how to repair the ONT **106a** or another device in the PON **100**. The diagnostic information may include a download rate at the ONT **106a**, an upload rate at the ONT **106a**, a light level metric at the ONT **106a**, a distance from the ONT **106a** to the OLT **102**, a temperature metric at the ONT **106a**, a PON utilization metric, an uplink metric for the ONT **106a**, or any other suitable diagnostic information related to the ONT **106a** and/or the PON **100**. The diagnostic information may also include any suitable diagnostic information from the OLT **102**, the FDH **116a**, the FDT, or any other optical terminal in the PON **100**. For example, the diagnostic information may further include diagnostic information from neighboring ONTs, so that the user can determine that a neighbor's ONT is the cause of the network failure. Then the user may contact the neighbor to repair the neighbor's ONT.

(63) In addition to presenting diagnostic information on the client device **150**, the PON repair application may generate and provide instructions for repairing the ONT **106a** based on the diagnostic information. More specifically, the PON repair application may receive the diagnostic information from the client device **150** and may analyze the diagnostic information to determine the root cause of the network failure. Additionally, the PON repair application may receive other sensor data indicating the current environment around the ONT **106a**. For example, the PON repair application may obtain images of the ONT **106a** from the camera. In some scenarios such as when the client device **150** is smart glasses or a VR headset, the camera view may be same as the user's view and the PON repair application may obtain real-world images of the user's current view of optical equipment. The PON repair application may also receive infrared data via an infrared sensor at the client device **150**. For example, the ONT **106a** may transmit IR signals that indicate various issues with the ONT **106a**.

(64) The PON repair application may then analyze the sensor data indicating the current environment at the ONT **106a** to determine the cause of the network failure. For example, the PON repair application may analyze the images to identify a blinking LED on the ONT. Each LED may signal a different issue and the LED may emit light with multiple blinking patterns or in multiple different colors which each signal a different issue. The PON repair application may then determine

the cause of the network failure based on which LEDs are lit up, which color each LED emits, and/or which blinking pattern each LED emits (e.g., 5 consecutive blinks, two blinks followed by a 5 second pause followed by 2 more blinks, etc.). The PON repair application may analyze the images using object recognition and/or optical character recognition (OCR) techniques to identify the LEDs that are lit up and/or their blinking patterns.

(65) More specifically, the PON repair application may analyze a portion (also referred to herein as an “object”) of an image and identify features of the object within the image, such as the geometry of the edges of the object, and RGB pixel values or colors within the object.

(66) These features may be identified by detecting stable regions within the object that are detectable regardless of blur, motion, distortion, orientation, illumination, scaling, and/or other changes in camera perspective. The stable regions may be extracted from the object using a scale-invariant feature transform (SIFT), speeded up robust features (SURF), fast retina keypoint (FREAK), binary robust invariant scalable keypoints (BRISK), or any other suitable computer vision techniques. In some embodiments, keypoints may be located at high-contrast regions of the object, such as edges within the object. A bounding box may be formed around a keypoint and the portion of the object created by the bounding box may be a feature.

(67) In any event, the PON repair application may compare the features identified for the object to features from template objects (also referred to herein as “template features”) using image classification and/or machine learning techniques, where the template objects may include optical terminals or components of optical terminals, such as LEDs on an optical terminal, ports, fibers connected to the optical terminal, etc. The machine learning techniques may include linear regression, polynomial regression, logistic regression, random forests, boosting, nearest neighbors, Bayesian networks, neural networks, support vector machines, or any other suitable machine learning technique. The template objects may also include other representations which are not of optical terminals. Then each of these template features may be compared to the features for an object.

(68) In some embodiments, the template features may be compared to the features for an object using a nearest neighbors algorithm. The nearest neighbors algorithm may identify template features which are the closest to the features of the object by creating numerical representations of the features to generate feature vectors, such as a pixel width and height of the object, and RGB pixel values for the object. The numerical representations of the features or feature vectors of the object may be compared to the feature vectors of template objects to determine a vector distance between the features of the object and each template object. The PON repair application may then identify the object based on the amount of similarity, or the vector distance in the nearest neighbors algorithm, between the features for the object and the features for template objects. If the closest template objects represent an LED, the object is identified as an LED. If the closest template objects represent an ONT, the object is identified as an ONT. Additionally, if the closest template objects represent a fiber, the object is identified as a fiber.

(69) In any event, the PON repair application may identify an LED in an image and identify the signal to which the LED corresponds based on the shape of the LED or based on text which is adjacent to the LED. For example, if the text is “Power,” the PON repair application may determine that the LED represents a power signal and if the LED is off, the PON repair application may determine that the power is off for the ONT **106a**. Additionally, if the text is “Service,” the PON repair application may determine that the LED represents an Internet service signal. If the LED is off, the PON repair application may determine that the ONT **106a** is not connected to an Internet service provider.

(70) In another example, the PON repair application may analyze the images to determine that a fiber is bent. For example, the PON repair application may identify the fiber using the objection recognition techniques described above. The PON repair application may then determine the geometry of the identified fiber and if the fiber curves at an angle which is more than a threshold

angle (e.g., 45 degrees), the PON repair application may determine that the fiber is bent. In yet another example, the PON repair application may determine that the fiber is dirty, for example based on the colors of the fiber or based on the ONT **106a** detecting Rayleigh scattering in the fiber according to attenuation in the detected light level. In another example, the PON repair application may determine that the fiber is broken by detecting cracks or breaks in the fiber.

(71) The PON repair application may also analyze the diagnostic information from the ONT **106a** and/or other sensor data to determine that the fiber is broken or bent. In other implementations, the PON repair application may analyze the diagnostic information and/or other sensor data to determine that the fiber is the cause of the network failure. Then the PON repair application may instruct the user to look at the fiber to determine whether the fiber is broken, bent, or dirty and to repair the ONT **106a** accordingly based on the user's findings.

(72) In addition to identifying issues with the fiber, the PON repair application may analyze the diagnostic information from the ONT **106a** and/or other sensor data to determine that the ONT **106a** needs to be replaced, that the ONT **106a** needs to be reset (e.g., by turning the power on and off), that an outlet powering the ONT **106a** needs to be reset, that an Ethernet cable connected to the ONT **106a** needs to be replaced, that an optical splitter needs to be replaced **114a**, etc.

(73) Then in response to determining the cause of the network failure, the PON repair application may generate and provide instructions for repairing the ONT **106a**. For example, when a fiber is bent, the PON repair application may generate instructions to straighten out the fiber. When the fiber is dirty, the PON repair application may generate instructions for cleaning the fiber. When the fiber is broken, the PON repair application may generate instructions for replacing the fiber with another fiber. In another example, when the ONT **106a** needs to be reset, the PON repair application may generate instructions for resetting the ONT **106a**. When the ONT **106a** is not receiving power from the outlet, the PON repair application may generate instructions for resetting the outlet to the ONT **106a**. Furthermore, when the ONT **106a** needs to be replaced, the PON repair application may generate instructions for removing the current ONT **106a** and installing a new ONT.

(74) While these are a few examples of the causes of network failures and corresponding instructions for repairing the network failures, the PON repair application may identify any suitable cause of a network failure and generate corresponding instructions for repairing the network failure according to the identified cause.

(75) In any event, the PON repair application may then present the instructions to the user for example, as step-by-step instructions as the user repairs the ONT **106a**. The PON repair application may present the instructions audibly via a speaker of the client device **150** or visually on a display. In some scenarios, the PON repair application may transition to an augmented reality mode that presents real-world imagery on a transparent display in the field of view of the client device **150**, and more specifically the camera view of the camera. The real-world imagery may be the user's view of the area in front of them so that they may view the instructions as they are repairing the ONT **106a**. In some implementations, the PON repair application may overlay the instructions on the transparent display in a display area within the transparent display that does not obstruct the user's view of the ONT **106a**. The PON repair application may identify the ONT **106a** within the real-world imagery using the object recognition techniques described above. Then the PON repair application may identify a display area within the real-world imagery that does not obstruct the ONT **106a**. For example, the display area may be above, below, to the left, or to the right of the ONT **106a**. The PON repair application may overlay the instructions on the real-world imagery within the identified display area.

(76) In this manner, when the client device **150** is smart glasses or a VR headset, the user may look at the ONT **106a** and see instructions for repairing the ONT **106a** in their field of view as they repair the ONT **106a**. When the client device **150** is a tablet or smart phone, the user may also be able to hold their tablet or smart phone up to the ONT **106a** and see the instructions for repairing

the ONT **106a**.

(77) In addition to overlaying text instructions on the real-world imagery in the augmented reality mode, the PON repair application may overlay augmented reality features on the real-world imagery to assist the user in repairing the ONT **106a** or other optical terminal, such as graphics or icons which highlight portions of the real-world imagery. For example, when the instructions are to examine a particular fiber, the PON repair application may identify the location of the particular fiber within the real-world imagery using the object recognition techniques described above. For example, the PON repair application may identify the fiber and identify the pixel coordinates of the fiber within the real-world imagery. Then the PON repair application may overlay a graphic at the identified location for the fiber to highlight the fiber relative to other objects within the image. For example, the graphic may be the same size and shape as the fiber in a bright yellow color, so that the fiber appears bright yellow and is easily noticeable by the user.

(78) More generally, the PON repair application may identify location(s) within the real-world imagery for the user to view and may overlay augmented reality feature(s) at those location(s). FIG. 7 illustrates an example transparent display **700** of a client device **150** with real-world imagery in the field of view of a user **10**, where the PON repair application is in an augmented reality mode. The real-world imagery includes the ONT **106a**. Additionally, the transparent display includes a set of step-by-step instructions **702** for repairing the ONT **106a** in a text format. The set of step-by-step instructions **702** are presented within a display area which does not obstruct the ONT **106a**. In this manner, the user may view and repair the ONT **106a** while viewing the instructions **702**. The real-world imagery also includes a power cord **706** connected to a power outlet for the ONT **106a**, and two optical fibers **708**, **710** connected to ports **4** and **8** of the ONT **106a**, respectively. The instructions **702** are to disconnect the fiber **708** from port **4** and replace it with a new fiber optic cable.

(79) In some implementations, the transparent display may present one instruction at a time and present a new instruction once the previous instruction has been completed. FIG. 8 illustrates an example transparent display **800** of the client device **150** with real-world imagery of the user's field of view. The transparent display **800** includes a first instruction **802** for repairing the ONT **106a** which is to remove the fiber optic cable from port **4**. To assist the user in identifying the fiber optic cable and port **4**, the PON repair application overlays a first augmented reality feature **804** over port **4** and a second augmented reality feature **806** over the fiber. While the first and second augmented reality features **804**, **806** are depicted as semi-transparent rectangular and oval shapes, respectively, the augmented reality features may be any suitable size, shape, or color to highlight or assist the user in viewing a particular portion of the real-world imagery.

(80) In any event, in response to determining that the first instruction **802** has been completed, for example by analyzing updated real-world imagery to determine that the fiber optic cable has been removed from port **4**, the transparent display **800** may present a second instruction **902** and/or overlay new AR features for the second instruction **902**. In other implementations, the user may select a user control to indicate that the instruction has been completed. FIG. 9 illustrates another example transparent display **900** of the client device **150** with updated real-world imagery of the user's field of view. The transparent display **900** includes a second instruction **902** for repairing the ONT **106a** which is to insert the new fiber optic cable into port **4**. To assist the user in identifying port **4**, the PON repair application overlays an augmented reality feature **904** over port **4**. The augmented reality feature **904** may be the same augmented reality feature **804** overlaid over port **4** for the first instruction **802** or may be a different augmented reality feature.

(81) FIG. 10 depicts a flow diagram of an example method **1000** for repairing a network failure in a PON **100**. The method **1000** may be performed by the client device **150**. At block **1002**, the client device **150** establishes a communication session with the ONT **106a** via a short-range communication link. For example, the client device **150** and the ONT **106a** may connect using Wi-Fi, Bluetooth, or NFC. More specifically, a user may tap the client device **150** to the ONT **106a** to

cause the PON repair application to transmit an NFC signal to the ONT **106a** requesting to pair the devices. The ONT **106a** may receive the NFC signal and establish a communication session between the two devices **150**, **106a** in response to the request. In another example, the ONT **106a** may broadcast a Wi-Fi signal. When the client device **150** is within communication range of the ONT **106a**, the PON repair application may receive the Wi-Fi signal and may request to connect to the ONT **106a** via the broadcasted Wi-Fi network.

(82) Then at block **1004**, in response to establishing the communication session, the client device **150** receives diagnostic information related to a network failure from the ONT **106a** via the short-range communication link. The diagnostic information may be real-time diagnostic information collected at the ONT **106a**. Additionally, the diagnostic information may include diagnostic information collected at the OLT **102**, FDH **116a**, or FDT and transmitted to the ONT **106a** via the fiber optic network or a Wi-Fi network. The diagnostic information may include a download rate at the ONT **106a**, an upload rate at the ONT **106a**, a light level metric at the ONT **106a**, a distance from the ONT **106a** to the OLT **102**, a temperature metric at the ONT **106a**, a PON utilization metric, an uplink metric for the ONT **106a**, etc.

(83) At block **1006**, the client device **150** presents the diagnostic information via a user interface. The user may then determine the cause of the network failure and/or repair the ONT **106a** using the diagnostic information.

(84) FIG. **11** depicts a flow diagram of an example method **1100** for augmenting repair of a network failure in a PON. The method **1000** may be performed by the client device **150**, the server **130**, or any suitable combination of the two. At block **1102**, the client device **150** receives sensor data indicative of the current environment at the ONT **106a**. The sensor data may include wireless communication signals from the ONT **106a** via a short-range communication network which include diagnostic information collected at the ONT **106a**. The diagnostic information may be diagnostic information determined by performing a diagnostic test, such as a dial tone test or a speed test. The sensor data may also include wireless communication signals from the FDH **116a**, FDT, or OLT **102** optically coupled to the ONT **106a**. Additionally, the sensor data may include infrared signals from the ONT **106a** and/or images of the ONT **106a**. The images may depict real-world imagery of the user's field of view or a camera view of the client device **150**.

(85) The client device **150** analyzes the current environment using the sensor data to determine the cause of a network failure related to the ONT **106a** (block **1104**). For example, the network failure may be due to the ONT **106a**, one or more fibers connected to the ONT **106a**, the FDH **116a**, the FDT, and/or the OLT. Some example network failure causes may include a dirty, bent, or broken fiber, a broken Ethernet cable, a lack of power to the ONT **106a**, a faulty ONT **106a** which may require a reset of the ONT **106a** or replacement, or another faulty piece of optical equipment, such as a faulty FDH, FDT, splitter, or a neighbor's ONT.

(86) Then at block **1106**, the client device **150** generates a set of instructions for repairing the ONT **106a** based on detected cause of the network failure. For example, when a fiber is bent, the client device **150** may generate instructions to straighten out the fiber. When the fiber is dirty, the client device **150** may generate instructions for cleaning the fiber. When the fiber is broken, the client device **150** may generate instructions for replacing the fiber with another fiber. In another example, when the ONT **106a** needs to be reset, the client device **150** may generate instructions for resetting the ONT **106a**. When the ONT **106a** is not receiving power from the outlet, the client device **150** may generate instructions for resetting the outlet to the ONT **106a**. Furthermore, when the ONT **106a** needs to be replaced, the client device **150** may generate instructions for removing the current ONT **106a** and installing a new ONT.

(87) The client device **150** may also analyze the sensor data to identify locations within real-world imagery to highlight when instructing the user on how to repair the ONT **106a** (block **1108**). For example, when the instructions are to examine a particular fiber, the client device **150** may identify the location of the particular fiber within the real-world imagery using the object recognition

techniques described above. For example, the client device **150** may identify the fiber and identify the pixel coordinates of the fiber within the real-world imagery.

(88) At block **1110**, the client device **150** provides the set of instructions for repairing the ONT **106a**. The client device **150** may present the instructions audibly via a speaker of the client device **150** or visually on a display. In some scenarios, the client device **150** may transition to an augmented reality mode that presents real-world imagery on a transparent display in the field of view of the client device **150**, and more specifically the camera view of the camera. The real-world imagery may be the user's view of the area in front of them so that they may view the instructions as they are repairing the ONT **106a**. In some implementations, the client device **150** may overlay the instructions on the transparent display in a display area within the transparent display that does not obstruct the user's view of the ONT **106a**.

(89) Additionally, the client device **150** provides augmented reality features overlaid on the real-world imagery at the identified locations to assist the user in repairing the ONT **106a** (block **1112**). The augmented reality features may be graphics or icons which highlight portions of the real-world imagery. For example, the client device **150** may overlay a graphic at the identified location for the fiber to highlight the fiber relative to other objects within the image. The graphic may be the same size and shape as the fiber in a bright yellow color, so that the fiber appears bright yellow and is easily noticeable by the user.

ADDITIONAL CONSIDERATIONS

(90) The above description refers to a block diagram of the accompanying drawings. Alternative implementations of the example represented by the block diagram includes one or more additional or alternative elements, processes and/or devices. Additionally or alternatively, one or more of the example blocks of the diagram may be combined, divided, re-arranged or omitted. Components represented by the blocks of the diagram are implemented by hardware, software, firmware, and/or any combination of hardware, software and/or firmware. In some examples, at least one of the components represented by the blocks is implemented by a logic circuit. As used herein, the term "logic circuit" is expressly defined as a physical device including at least one hardware component configured (e.g., via operation in accordance with a predetermined configuration and/or via execution of stored machine-readable instructions) to control one or more machines and/or perform operations of one or more machines. Examples of a logic circuit include one or more processors, one or more coprocessors, one or more microprocessors, one or more controllers, one or more digital signal processors (DSPs), one or more application specific integrated circuits (ASICs), one or more field programmable gate arrays (FPGAs), one or more microcontroller units (MCUs), one or more hardware accelerators, one or more special-purpose computer chips, and one or more system-on-a-chip (SoC) devices. Some example logic circuits, such as ASICs or FPGAs, are specifically configured hardware for performing operations (e.g., one or more of the operations described herein and represented by the flowcharts of this disclosure, if such are present). Some example logic circuits are hardware that executes machine-readable instructions to perform operations (e.g., one or more of the operations described herein and represented by the flowcharts of this disclosure, if such are present). Some example logic circuits include a combination of specifically configured hardware and hardware that executes machine-readable instructions. The above description refers to various operations described herein and flowcharts that may be appended hereto to illustrate the flow of those operations. Any such flowcharts are representative of example methods disclosed herein. In some examples, the methods represented by the flowcharts implement the apparatus represented by the block diagrams. Alternative implementations of example methods disclosed herein may include additional or alternative operations. Further, operations of alternative implementations of the methods disclosed herein may combined, divided, re-arranged or omitted. In some examples, the operations described herein are implemented by machine-readable instructions (e.g., software and/or firmware) stored on a medium (e.g., a tangible machine-readable medium) for execution by one or more logic circuits (e.g., processor(s)). In some

examples, the operations described herein are implemented by one or more configurations of one or more specifically designed logic circuits (e.g., ASIC(s)). In some examples the operations described herein are implemented by a combination of specifically designed logic circuit(s) and machine-readable instructions stored on a medium (e.g., a tangible machine-readable medium) for execution by logic circuit(s).

(91) As used herein, each of the terms “tangible machine-readable medium,” “non-transitory machine-readable medium” and “machine-readable storage device” is expressly defined as a storage medium (e.g., a platter of a hard disk drive, a digital versatile disc, a compact disc, flash memory, read-only memory, random-access memory, etc.) on which machine-readable instructions (e.g., program code in the form of, for example, software and/or firmware) are stored for any suitable duration of time (e.g., permanently, for an extended period of time (e.g., while a program associated with the machine-readable instructions is executing), and/or a short period of time (e.g., while the machine-readable instructions are cached and/or during a buffering process)). Further, as used herein, each of the terms “tangible machine-readable medium,” “non-transitory machine-readable medium” and “machine-readable storage device” is expressly defined to exclude propagating signals. That is, as used in any claim of this patent, none of the terms “tangible machine-readable medium,” “non-transitory machine-readable medium,” and “machine-readable storage device” can be read to be implemented by a propagating signal.

(92) In the foregoing specification, specific examples have been described. However, one of ordinary skill in the art appreciates that various modifications and changes can be made without departing from the scope of the disclosure as set forth in the claims below. Accordingly, the specification and figures are to be regarded in an illustrative rather than a restrictive sense, and all such modifications are intended to be included within the scope of the disclosure. Additionally, the described examples should not be interpreted as mutually exclusive, and should instead be understood as potentially combinable if such combinations are permissive in any way. In other words, any feature disclosed in any of the aforementioned examples may be included in any of the other aforementioned examples.

(93) The benefits, advantages, solutions to problems, and any element(s) that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as a critical, required, or essential features or elements of any or all the claims. The claimed invention is defined solely by the appended claims including any amendments made during the pendency of this application and all equivalents of those claims as issued.

(94) Moreover in this document, relational terms such as first and second, top and bottom, and the like may be used solely to distinguish one entity or action from another entity or action without necessarily requiring or implying any actual such relationship or order between such entities or actions. The terms “comprises,” “comprising,” “has,” “having,” “includes,” “including,” “contains,” “containing” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises, has, includes, contains a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element preceded by “comprises . . . a”, “has . . . a”, “includes . . . a”, “contains . . . a” does not, without more constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises, has, includes, contains the element. The terms “a” and “an” are defined as one or more unless explicitly stated otherwise herein. The terms “substantially”, “essentially”, “approximately”, “about” or any other version thereof, are defined as being close to as understood by one of ordinary skill in the art, and in one non-limiting example the term is defined to be within 10%, in another example within 5%, in another example within 1% and in another example within 0.5%. The term “coupled” as used herein is defined as connected, although not necessarily directly and not necessarily mechanically. A device or structure that is “configured” in a certain way is configured in at least that way, but may also be configured in ways that are not

listed.

(95) Further, unless expressly stated to the contrary, “or” refers to an inclusive or and not to an exclusive or. For example, “A, B or C” refers to any combination or subset of A, B, C such as (1) A alone, (2) B alone, (3) C alone, (4) A with B, (5) A with C, (6) B with C, and (7) A with B and with C. As used herein, the phrase “at least one of A and B” is intended to refer to any combination or subset of A and B such as (1) at least one A, (2) at least one B, and (3) at least one A and at least one B. Similarly, the phrase “at least one of A or B” is intended to refer to any combination or subset of A and B such as (1) at least one A, (2) at least one B, and (3) at least one A and at least one B.

(96) The Abstract of the Disclosure is provided to allow the reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, it can be seen that various features are grouped together in various examples for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed examples require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter may lie in less than all features of a single disclosed example. Thus, the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separately claimed subject matter.

(97) Further, as used herein, the expressions “in communication,” “coupled” and “connected,” including variations thereof, encompasses direct communication and/or indirect communication through one or more intermediary components, and does not require direct mechanical or physical (e.g., wired) communication and/or constant communication, but rather additionally includes selective communication at periodic intervals, scheduled intervals, aperiodic intervals, and/or one-time events. The examples are not limited in this context.

(98) Further still, unless expressly stated to the contrary, “or” refers to an inclusive or and not to an exclusive or. For example, “A, B or C” refers to any combination or subset of A, B, C such as (1) A alone, (2) B alone, (3) C alone, (4) A with B, (5) A with C, (6) B with C, and (7) A with B and with C. As used herein, the phrase “at least one of A and B” is intended to refer to any combination or subset of A and B such as (1) at least one A, (2) at least one B, and (3) at least one A and at least one B. Similarly, the phrase “at least one of A or B” is intended to refer to any combination or subset of A and B such as (1) at least one A, (2) at least one B, and (3) at least one A and at least one B.

(99) Moreover, in the foregoing specification and the attached drawings, specific examples have been described. However, one of ordinary skill in the art appreciates that various modifications and changes can be made in view of aspects of this disclosure without departing from the scope of the disclosure as set forth in the claims below. Accordingly, the specification and figures are to be regarded in an illustrative rather than a restrictive sense, and all such modifications made in view of aspects of this disclosure are intended to be included within the scope of present teachings. Numerous alternative examples could be implemented, using either current technology or technology developed after the filing date of this patent, which would still fall within the scope of the claims. By way of example, and not limitation, the disclosure herein contemplates at least the following examples:

(100) 1A. A method for repairing a network failure in a Passive Optical Network (PON), the method comprising: establishing a communication session, by a client device via a short-range communication link, with an optical network terminal (ONT) in a PON; in response to establishing the communication session, receiving, at the client device from the ONT via the short-range communication link, diagnostic information related to a network failure corresponding to the ONT; and presenting, by the client device, the diagnostic information to a user for the user to repair the ONT based on the received diagnostic information.

(101) 2A. The method of example 1A, further comprising: transmitting, by the client device to the

ONT via the short-range communication link, a request for the ONT to perform a diagnostic test; and receiving, at the client device from the ONT via the short-range communication link, the diagnostic information in response to the ONT performing the diagnostic test.

(102) 3A. The method of any of the preceding examples, wherein the diagnostic test includes at least one of: a dial tone test to test whether the ONT can successfully complete a call with the client device or a speed test to test download or upload rates at the ONT.

(103) 4A. The method of any of the preceding examples, wherein the short-range communication link is at least one of: a Wi-Fi communication link, a near field communication (NFC) link, or a Bluetooth communication link.

(104) 5A. The method of any of the preceding examples, wherein establishing the communication session includes: receiving, at the client device, a signal from the ONT broadcasted via the Wi-Fi communication link; transmitting, by the client device via the Wi-Fi communication link, a request to connect to the ONT in response to receiving the signal, wherein the communication session is established in response to the ONT receiving the request.

(105) 6A. The method of any of the preceding examples, wherein establishing the communication session includes: transmitting, by the client device via the NFC link, a request to pair the client device with the ONT, wherein the communication session is established in response to the ONT receiving the request.

(106) 7A. The method of any of the preceding examples, wherein establishing the communication session includes: receiving, at the client device, a search query including identification information for the ONT; identifying, by the client device, the ONT based on the search query; detecting, by the client device, a short-range communication signal from the identified ONT; and transmitting, by the client device, a request to connect to the ONT in response to receiving the short-range communication signal.

(107) 8A. The method of any of the preceding examples, wherein the diagnostic information includes at least one of: a download rate at the ONT, an upload rate at the ONT, a light level metric at the ONT, a distance from the ONT to an optical line terminal (OLT), a temperature metric at the ONT, a PON utilization metric, or an uplink metric for the ONT.

(108) 9A. A client device for repairing a network failure in a Passive Optical Network (PON), the client device comprising: one or more processors; a non-transitory computer-readable memory storing instructions thereon that, when executed by the one or more processors, cause the client device to: establish, via a short-range communication link, a communication session with an optical network terminal (ONT) in a PON; in response to establishing the communication session, receive, from the ONT via the short-range communication link, diagnostic information related to a network failure corresponding to the ONT; and present the diagnostic information to a user for the user to repair the ONT based on the received diagnostic information.

(109) 10A. The client device of example 9A, wherein the instructions further cause the client device to: transmit, to the ONT via the short-range communication link, a request for the ONT to perform a diagnostic test; and receive, from the ONT via the short-range communication link, the diagnostic information in response to the ONT performing the diagnostic test.

(110) 11A. The client device of example 9A or example 10A, wherein the diagnostic test includes at least one of: a dial tone test to test whether the ONT can successfully complete a call with the client device or a speed test to test download or upload rates at the ONT.

(111) 12A. The client device of any of examples 9A to 11A, wherein the short-range communication link is at least one of: a Wi-Fi communication link, a near field communication (NFC) link, or a Bluetooth communication link.

(112) 13A. The client device of any of examples 9A to 12A, wherein to establish the communication session, the instructions cause the client device to: receive a signal from the ONT broadcasted via the Wi-Fi communication link; and transmit, via the Wi-Fi communication link, a request to connect to the ONT in response to receiving the signal, wherein the communication

session is established in response to the ONT receiving the request.

(113) 14A. The client device of any of examples 9A to 13A, wherein to establish the communication session, the instructions cause the client device to: transmit, via the NFC link, a request to pair the client device with the ONT, wherein the communication session is established in response to the ONT receiving the request.

(114) 15A. The client device of any of examples 9A to 14A, wherein to establish the communication session, the instructions cause the client device to: receive a search query including identification information for the ONT; identify the ONT based on the search query; detect a short-range communication signal from the identified ONT; and transmit a request to connect to the ONT in response to receiving the short-range communication signal.

(115) 16A. The client device of any of examples 9A to 15A, wherein the diagnostic information includes at least one of: a download rate at the ONT, an upload rate at the ONT, a light level metric at the ONT, a distance from the ONT to an optical line terminal (OLT), a temperature metric at the ONT, a PON utilization metric, or an uplink metric for the ONT.

(116) 17A. A non-transitory computer-readable memory storing instructions thereon that, when executed by one or more processors, cause the one or more processors to: establish, via a short-range communication link, a communication session with an optical network terminal (ONT) in a PON; in response to establishing the communication session, receive, from the ONT via the short-range communication link, diagnostic information related to a network failure corresponding to the ONT; and present the diagnostic information to a user for the user to repair the ONT based on the received diagnostic information.

(117) 18A. The non-transitory computer-readable memory of example 17A, wherein the instructions further cause the one or more processors to: transmit, to the ONT via the short-range communication link, a request for the ONT to perform a diagnostic test; and receive, from the ONT via the short-range communication link, the diagnostic information in response to the ONT performing the diagnostic test.

(118) 19A. The non-transitory computer-readable memory of example 17A or example 18A, wherein the diagnostic test includes at least one of: a dial tone test to test whether the ONT can successfully complete a call with the client device or a speed test to test download or upload rates at the ONT.

(119) 20A. The non-transitory computer-readable memory of any of examples 17A to 19A, wherein the short-range communication link is at least one of: a Wi-Fi communication link, a near field communication (NFC) link, or a Bluetooth communication link.

(120) 1B. A method for augmenting repair of a network failure in a Passive Optical Network (PON), the method comprising: receiving, via one or more sensors in a client device, sensor data indicating a current environment that includes an optical network terminal (ONT) in a PON, the PON including an optical line terminal (OLT) optically connected to the ONT via one or more optical fibers; analyzing, by the client device, the current environment to detect a cause of a network failure corresponding to the ONT; generating, by the client device, a set of instructions for repairing the ONT based on the detected cause of the network failure; and providing, by the client device, the set of instructions for a user to follow to repair the ONT, wherein the set of instructions are provided as the user repairs the ONT.

(121) 2B. The method of example 1B, wherein providing the set of instructions includes: providing, by the client device, the set of instructions via a transparent display overlaying the user's field of view, wherein the ONT is within the user's field of view in the transparent display and the set of instructions are presented within a display area which does not obstruct the ONT.

(122) 3B. The method of example 1B or example 2B, wherein providing the set of instructions includes providing, by the client device, a set of audio instructions via a speaker.

(123) 4B. The method of any of example 1B to 3B, further comprising: analyzing, by the client device, the current environment to identify one or more locations for the user to view to repair the

ONT; and providing, by the client device via a transparent display overlaying the user's field of view, one or more augmented reality features overlaid on the user's real-world view of the current environment at the identified one or more locations which assist the user in repairing the ONT.

(124) 5B. The method of any of example 1B to 4B, wherein the receiving sensor data indicating the current environment includes receiving, at the client device, via the one or more sensors, at least one of: one or more images depicting a field of view of the user which includes the ONT, one or more infrared signals from the ONT, one or more wireless communication signals from the ONT, or one or more wireless communication signals from a fiber distribution hub (FDH) or fiber distribution terminal (FDT) optically coupled to the ONT.

(125) 6B. The method of any of example 1B to 5B, wherein the one or more wireless communication signals from the ONT include diagnostic information for the ONT determined by performing one or more diagnostic tests.

(126) 7B. The method of any of example 1B to 6B, wherein the set of instructions for repairing the ONT are generated based on the diagnostic information.

(127) 8B. A client device for augmenting repair of a network failure in a Passive Optical Network (PON), the client device comprising: one or more sensors; one or more processors; a non-transitory computer-readable memory storing instructions thereon that, when executed by the one or more processors, cause the client device to: receive, via the one or more sensors, sensor data indicating a current environment including an optical network terminal (ONT) in a PON, the PON including an optical line terminal (OLT) optically connected to the ONT via one or more optical fibers; analyze the current environment to generate a set of instructions for repairing the ONT; and provide the set of instructions for a user to follow to repair the ONT, wherein the set of instructions are provided as the user repairs the ONT.

(128) 9B. The client device of example 8B, further comprising a transparent display overlaying the user's field of view, wherein the set of instructions are provided via the transparent display, and wherein the ONT is within the user's field of view in the transparent display and the set of instructions are presented within a display area which does not obstruct the ONT.

(129) 10B. The client device of example 8B or example 9B, further comprising a speaker, wherein the set of instructions are provided as a set of audio instructions via the speaker.

(130) 11B. The client device of any of examples 8B to 10B, further comprising a transparent display overlaying the user's field of view, wherein the instructions further cause the client device to: analyze the current environment to identify one or more locations for the user to view to repair the ONT; and provide, via the transparent display, one or more augmented reality features overlaid on the user's real-world view of the current environment at the identified one or more locations which assist the user in repairing the ONT.

(131) 12B. The client device of any of examples 8B to 11B, wherein the sensor data includes at least one of: one or more images depicting a field of view of the user which includes the ONT, one or more electromagnetic signals at the client device, one or more infrared signals from the ONT, one or more wireless communication signals from the ONT, or one or more wireless communication signals from a fiber distribution hub (FDH) or fiber distribution terminal (FDT) optically coupled to the ONT.

(132) 13B. The client device of any of examples 8B to 12B, wherein the one or more wireless communication signals from the ONT include diagnostic information for the ONT determined by performing one or more diagnostic tests.

(133) 14B. The client device of any of examples 8B to 13B, wherein the set of instructions for repairing the ONT are generated based on the diagnostic information.

(134) 15B. A non-transitory computer-readable memory storing instructions thereon that, when executed by one or more processors, cause the one or more processors to: receive, via one or more sensors, sensor data indicating a current environment including an optical network terminal (ONT) in a PON, the PON including an optical line terminal (OLT) optically connected to the ONT via

one or more optical fibers; analyze the current environment to generate a set of instructions for repairing the ONT; and provide the set of instructions for a user to follow to repair the ONT, wherein the set of instructions are provided as the user repairs the ONT.

(135) 16B. The non-transitory computer-readable memory of example 15B, wherein the set of instructions are provided via a transparent display, and wherein the ONT is within the user's field of view in the transparent display and the set of instructions are presented within a display area which does not obstruct the ONT.

(136) 17B. The non-transitory computer-readable memory of example 15B or example 16B, wherein the set of instructions are provided as a set of audio instructions via the speaker.

(137) 18B. The non-transitory computer-readable memory of any of examples 15B to 17B, wherein the instructions further cause the one or more processors to: analyze the current environment to identify one or more locations for the user to view to repair the ONT; and provide, via a transparent display, one or more augmented reality features overlaid on the user's real-world view of the current environment at the identified one or more locations which assist the user in repairing the ONT.

(138) 19B. The non-transitory computer-readable memory of any of examples 15B to 18B, wherein the sensor data includes at least one of: one or more images depicting a field of view of the user which includes the ONT, one or more electromagnetic signals at the client device, one or more infrared signals from the ONT, one or more wireless communication signals from the ONT, or one or more wireless communication signals from a fiber distribution hub (FDH) or fiber distribution terminal (FDT) optically coupled to the ONT.

(139) 20B. The non-transitory computer-readable memory of any of examples 15B to 19B, wherein the one or more wireless communication signals from the ONT include diagnostic information for the ONT determined by performing one or more diagnostic tests.

(140) Additionally, the benefits, advantages, solutions to problems, and any element(s) that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as a critical, required, or essential features or elements of any or all the claims.

(141) Finally, any references, including, but not limited to, publications, patent applications, and patents cited herein are hereby incorporated in their entirety by reference to the same extent as if each reference were individually and specifically indicated to be incorporated by reference and were set forth in its entirety herein.

(142) The patent claims at the end of this patent application are not intended to be construed under 35 U.S.C. § 112(f) unless traditional means-plus-function language is expressly recited, such as “means for” or “step for” language being explicitly recited in the claim(s). The systems and methods described herein are directed to an improvement to computer functionality, and improve the functioning of conventional computers.

(143) Although certain example methods, apparatus and articles of manufacture have been disclosed herein, the scope of coverage of this patent is not limited thereto. On the contrary, this patent covers all methods, apparatus and articles of manufacture fairly falling within the scope of the claims of this patent.

Claims

1. A method for repairing a network failure in a Passive Optical Network (PON), the method comprising: establishing a communication session, by a client device via a short-range communication link, with an optical network terminal (ONT) in a PON; in response to establishing the communication session, transmitting, by the client device to the ONT via the short-range communication link, a request for the ONT to perform a dial tone test to test whether the ONT can successfully complete a call with the client device via the PON and a cellular or radio network; and generating diagnostic information indicating whether the call is received at the client device from

the ONT via the PON in response to the request to perform the dial tone test.

2. The method of claim 1, further comprising: transmitting, by the client device to the ONT via the short-range communication link, a request for the ONT to perform a speed test to test download or upload rates at the ONT; and generating additional diagnostic information in response to the ONT performing the speed test.

3. The method of claim 1, wherein the short-range communication link is at least one of: a Wi-Fi communication link, or a near field communication (NFC) link.

4. The method of claim 3, wherein establishing the communication session includes: receiving, at the client device, a signal from the ONT broadcasted via the Wi-Fi communication link; transmitting, by the client device via the Wi-Fi communication link, a request to connect to the ONT in response to receiving the signal, wherein the communication session is established in response to the ONT receiving the request.

5. The method of claim 3, wherein establishing the communication session includes: transmitting, by the client device via the NFC link, a request to pair the client device with the ONT, wherein the communication session is established in response to the ONT receiving the request.

6. The method of claim 3, wherein establishing the communication session includes: receiving, at the client device, a search query including identification information for the ONT; identifying, by the client device, the ONT based on the search query; identifying, by the client device, a network name for connecting to the ONT; presenting, by the client device, the network name for connecting to the ONT; detecting, by the client device, a short-range communication signal from the identified ONT; and transmitting, by the client device, a request to connect to the ONT via the network in response to receiving the short-range communication signal.

7. The method of claim 1, further comprising: presenting, by the client device, the diagnostic information to a user for the user to repair the ONT based on the diagnostic information.

8. The method of claim 1, further comprising: presenting, via a user interface of the client device, a user control for initiating the dial tone test which, when selected, causes the client device to transmit the request for the ONT to perform the dial tone test; and in response to receiving the call from the ONT, presenting, via the user interface of the client device, an indication of the call being received from the ONT.

9. A client device for repairing a network failure in a Passive Optical Network (PON), the client device comprising: one or more processors; a non-transitory computer-readable memory storing instructions thereon that, when executed by the one or more processors, cause the client device to: establish, via a short-range communication link, a communication session with an optical network terminal (ONT) in a PON; in response to establishing the communication session, transmit, to the ONT via the short-range communication link, a request for the ONT to perform a dial tone test to test whether the ONT can successfully complete a call with the client device via the PON and a cellular or radio network; and generate diagnostic information indicating whether the call is received at the client device from the ONT via the PON in response to the request to perform the dial tone test.

10. The client device of claim 9, wherein the instructions further cause the client device to: transmit, to the ONT via the short-range communication link, a request for the ONT to perform a speed test to test download or upload rates at the ONT; and generate additional diagnostic information in response to the ONT performing the speed test.

11. The client device of claim 9, wherein the short-range communication link is at least one of: a Wi-Fi communication link, or a near field communication (NFC) link.

12. The client device of claim 11, wherein to establish the communication session, the instructions cause the client device to: receive a signal from the ONT broadcasted via the Wi-Fi communication link; and transmit, via the Wi-Fi communication link, a request to connect to the ONT in response to receiving the signal, wherein the communication session is established in response to the ONT receiving the request.

13. The client device of claim 11, wherein to establish the communication session, the instructions cause the client device to: transmit, via the NFC link, a request to pair the client device with the ONT, wherein the communication session is established in response to the ONT receiving the request.
 14. The client device of claim 11, wherein to establish the communication session, the instructions cause the client device to: receive a search query including identification information for the ONT; identify the ONT based on the search query; identify a network name for connecting to the ONT; present the network name for connecting to the ONT; detect a short-range communication signal from the identified ONT; and transmit a request to connect to the ONT via the network in response to receiving the short-range communication signal.
 15. The client device of claim 9, wherein the instructions further cause the client device to: present, via a user interface, a user control for initiating the dial tone test which, when selected, causes the client device to transmit the request for the ONT to perform the dial tone test; and in response to receiving the call from the ONT, present, via the user interface, an indication of the call being received from the ONT.
 16. A non-transitory computer-readable memory storing instructions thereon that, when executed by one or more processors, cause the one or more processors to: establish, via a short-range communication link, a communication session with an optical network terminal (ONT) in a PON; in response to establishing the communication session, transmit, to the ONT via the short-range communication link, a request for the ONT to perform a dial tone test to test whether the ONT can successfully complete a call with the client device via the PON and a cellular or radio network; and generate diagnostic information indicating whether the call is received at the client device from the ONT via the PON in response to the request to perform the dial tone test.
 17. The non-transitory computer-readable memory of claim 16, wherein the instructions further cause the one or more processors to: transmit, to the ONT via the short-range communication link, a request for the ONT to perform a speed test to test download or upload rates at the ONT; and generate additional diagnostic information in response to the ONT performing the speed test.
 18. The non-transitory computer-readable memory of claim 16, wherein the short-range communication link is at least one of: a Wi-Fi communication link or a near field communication (NFC) link.
 19. The non-transitory computer-readable memory of claim 18, wherein to establish the communication session, the instructions cause the one or more processors to: receive a signal from the ONT broadcasted via the Wi-Fi communication link; and transmit, via the Wi-Fi communication link, a request to connect to the ONT in response to receiving the signal, wherein the communication session is established in response to the ONT receiving the request.
 20. The non-transitory computer-readable memory of claim 18, wherein to establish the communication session, the instructions cause the one or more processors to: transmit, via the NFC link, a request to pair the client device with the ONT, wherein the communication session is established in response to the ONT receiving the request.
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